Christopher Wang

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Education

Massachusetts Institute of Technology — Cambridge, MA
B.S. in Computer Science & Mathematics, Double Major — *Aug. 2026 – May 2029 (expected)*
Coursework: Probability (18.600), Linear Algebra (18.C06), Programming (6.1010), C & Assembly (6.1903)

Solon High School — Solon, OH
4.00 Unweighted GPA — *Aug. 2022 – May 2026*

Awards

USNCO Camper: Placed amongst the top 20 on the US National Chemistry Olympiad exam in 2025, thus participating in the selection camp for IChO.

Experience

UROP — Sept. 2026 – Now
Massachusetts Institute of Technology — *Cambridge, MA*
- Developing an agentic pipeline that automatically extracts protein interaction data from scientific literature for phase separation research

Research Intern — June 2024 – Aug. 2026
Cleveland Clinic, Imaging Institute — *Cleveland, OH*
- Designed and implemented a Python image-processing pipeline to quantify hand tremor from patient handwriting samples, converting a clinical symptom into a measurable, reproducible signal.
- Built and shipped a prototype Android application for real-time monitoring of patient motor progress during HIFU thrombolysis treatment.
- Learned deep learning from scratch to extend the project: trained and validated a PyTorch model that replaced the deterministic algorithm with a data-driven tremor-severity classifier.
- Built the model evaluation pipeline (preprocessing, validation, metrics reporting) now in active use for ongoing HIFU thrombolysis clinical trials.

Garcia Research Scholar — June 2025 – Aug. 2025
Stony Brook University — *Stony Brook, NY*
- Designed a thermal shape-memory polymer for mechanical severance in brain aneurysm treatment, optimizing chemical composition against NMR/FTIR data to exceed 90% crosslinking efficiency.
- Co-authored a publication in MRS Communications and presented results at MRS Fall 2025.

Project Manager — Jan. 2024 – Dec. 2024
XCel Future — *Cleveland, OH*
- Planned and led events across two senior living facilities serving the elderly Asian community in Cleveland, and coordinated medical appointments to improve healthcare access.

Projects

NFL Prediction-Market Data Platform | *Python, asyncio, WebSockets, SQLite, pytest* — Sept. 2026 – Now
- Built an async Python platform (WebSockets, Ed25519-signed API auth, SQLite/zstd) that records live NFL play-by-play and Polymarket US order books: ∼300 markets per game, 15M updates across 13 games.
- Verified the platform's contract-payout logic against 4,460 settled markets with zero mismatches.
- Showed that free ESPN data trails market prices by a median 32 s (58 plays, 11 games, game-clustered bootstrap CI); traced an earlier, much smaller estimate to a 30 s cache in the exchange's public REST API.

Arithmetic Speed Drill | *JavaScript, HTML/CSS, Service Workers* | GitHub — Oct. 2026
- Built a Zetamac-style mental-math trainer as a dependency-free, installable iOS web app with configurable ranges.
- Cache-first service worker for full offline play; low-latency keypad; score history kept per settings combination.

Technical Skills

Languages: Python, Java, JavaScript, HTML/CSS
ML / Data Libraries: PyTorch, scikit-learn, NumPy, Pillow, Matplotlib
Developer Tools: Git, Linux/Unix, pytest, SQLite, VS Code, Visual Studio